Dylan Lee

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EDUCATION

University of California, Santa Barbara
B.S. in Computer Engineering Expected June 2029
GPA: 3.6/4.0
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), Computer Organization, Foundations of Computer Science, Analog & Digital Circuits and Systems

PROJECTS

LectureRecorder
Swift, SwiftUI, AVFoundation, Whisper, Metal

- Built a native macOS lecture recorder with AVAudioEngine and durable audio chunks, decoupling capture from transcription
- Integrated local Whisper transcription through a Swift/C worker with persistent state, process control, and failure handling
- Accelerated local Whisper inference with Metal and CPU fallback while isolating recording, storage, and transcription components
- Built automated tests covering recording lifecycle, audio capture, filesystem durability, process execution, and transcription failures

AI Voice Assistant
Python, asyncio, Whisper, Ollama, Vulkan, Kokoro

- Built a fully local voice assistant combining wake-word detection, VAD, Whisper, Ollama, Kokoro TTS, and real-time barge-in
- Offloaded speech recognition and LLM inference to the GPU with Vulkan while keeping wake-word detection and VAD on the CPU
- Used asyncio and dedicated worker executors to isolate blocking inference and synthesis from responsive microphone input
- Implemented concurrent VAD-based interruption to stop playback when new speech is detected during response generation

Job Application Autofill Agent
Python, Playwright, Ollama, Llama 3.1

- Built a Playwright pipeline to extract application fields, map candidate data with a local Ollama LLM, and populate forms
- Added confidence-based validation and a mandatory human review gate to flag uncertain responses before submission
- Built a resumable multi-job state machine with authenticated sessions, application logging, and recovery from navigation errors
- Handled text, select, checkbox, radio, file, and date controls with retry-aware Playwright filling

Budgeting Web Application
JavaScript, HTML/CSS, Firebase Auth, Firestore, Google Charts

- Built a budgeting web app with Firebase Authentication, Firestore persistence, and customizable expense categories
- Visualized income and spending with Google Charts and calculated daily and weekly spending targets

TECHNICAL SKILLS

Languages: Python, Swift, C++, Java, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: SwiftUI, AVFoundation, React, Next.js, Playwright
Tools & Technologies: Git, Firebase, Whisper, Ollama, Metal, Vulkan

EXPERIENCE

The Coder School Berkeley
Counselor June 2023 – July 2024

- Mentored groups of up to 15 students ages 6–12 in programming and project development
- Taught Python, Micro, Scratch, and introductory object-oriented programming concepts
- Developed individualized lessons to reinforce programming fundamentals, critical thinking, and creativity
- Guided students in designing and building original programming projects

Hanabi Judo
Assistant Coach October 2022 – August 2023

- Taught fundamental judo techniques to developing students during group practices
- Provided one-on-one coaching to help students improve technique and execution
- Assisted the head coach in planning and coordinating practice activities